Unit–V:

Data Interpretation: Tabulation, Bar Graph, Pie Chart, Line graph, Problem on data Data Interpretation: Sum and Difference based, Average based questions, percentage based questions.

Data Interpretation: Tabulation

This section comprises of questions in which certain data regarding common disciplines as production over a period of a few years : imports, exports, incomes of employees in a factory, students applying for and qualifying a certain field of study etc. are given in the form of a table. The candidate is required to understand the given information and thereafter answer the given questions on the basis of comparative analysis of the data.

Thus, here the data collected by the investigator are arranged in a systematic form in a table called the *tabular form*. In order to avoid some heads again and again, tables are made consisting of horizontal lines called *rows* and vertical lines called *columns* with distinctive heads, known as *captions*. Units of measurements are given with the captions.

Ex. 1. *The following table gives the sales of batteries manufactured by a company over the years. Study the table and answer the questions that follow :*

(S.B.I.P.O. 1998)

NUMBER OF DIFFERENT TYPES OF BATTERIES SOLD BY A COMPANY OVER THE YEARS (NUMBERS IN THOUSANDS)

Year	TYPES OF BATTERIES					
	4AH	7AH	32AH	35AH	55AH	Total
1992	75	144	114	102	108	543
1993	90	126	102	84	126	528
1994	96	114	75	105	135	525
1995	105	90	150	90	75	510
1996	90	75	135	75	90	465
1997	105	60	165	45	120	495
1998	115	85	160	100	145	605

1. The total sales of all the seven years is the maximum for which battery ?
   (a) 4AH (b) 7AH (c) 32AH (d) 35AH (e) 55AH
2. What is the difference in the number of 35AH batteries sold in 1993 and 1997 ?
   (a) 24000 (b) 28000 (c) 35000 (d) 39000 (e) 42000
3. The percentage of 4AH batteries sold to the total number of batteries sold was maximum in the year :
   (a) 1994 (b) 1995 (c) 1996 (d) 1997 (e) 1998
4. In the case of which battery there was a continuous decrease in sales from 1992 to 1997 ?
   (a) 4AH (b) 7AH (c) 32AH (d) 35AH (e) 55AH
5. What was the approximate percentage increase in the sales of 55AH batteries in 1998 compared to that in 1992 ?
   (a) 28% (b) 31% (c) 33% (d) 34% (e) 37%

Sol. 1. (*c*) : The total sales (in thousands) of all the seven years for various batteries are :

For 4AH = 75 + 90 + 96 + 105 + 90 + 105 + 115 = 676

For 7AH = 144 + 126 + 114 + 90 + 75 + 60 + 85 = 694

For 32AH = 114 + 102 + 75 + 150 + 135 + 165 + 160 = 901

For 35 AH = 102 + 84 + 105 + 90 + 75 + 45 + 100 = 601

For 55 AH = 108 + 126 + 135 + 75 + 90 + 120 + 145 = 799.

Clearly, sales are maximum in case of 32AH batteries.

2. (*d*) : Required difference = [(84 − 45) × 1000] = 39000.

3. (*d*) : The percentages of sales of 4AH batteries to the total sales in different years are :

For 1992 = $\left(\frac{75}{543} \times 100\right)\% = 13.81\%$; **For 1993** = $\left(\frac{90}{528} \times 100\right)\% = 17.05\%$;

For 1994 = $\left(\frac{96}{525} \times 100\right)\% = 18.29\%$; **For 1995** = $\left(\frac{105}{510} \times 100\right)\% = 20.59\%$;

For 1996 = $\left(\frac{96}{465} \times 100\right)\% = 19.35\%$; **For 1997** = $\left(\frac{105}{495} \times 100\right)\% = 21.21\%$;

For 1998 = $\left(\frac{115}{605} \times 100\right)\% = 19.01\%$.

Clearly, the percentage is maximum in 1997.

4. (*b*) : From the table it is clear that the sales of 7AH batteries have been decreasing continuously from 1992 to 1997.

5. (*d*) : Required Percentage = $\left[\frac{(145 - 108)}{108} \times 100\right]\% = 34.26\% = 34\%$.

Ex. 2. ***Study the follow***

What is Data Interpretation?

Data Interpretation refers to the extraction of data from the source. It can be obtained from multiple sources e.g. data from running of industries, census population data, etc. Data Interpretation basically requires the data first and analyzing data later to infer information from it in order to answer questions. Data can be provided in any format like Bars, tables, line graphs, pie graphs.

Types of Data Interpretation

Given below are the DI students are expected to comprehend data given and find out the answers:

- Tabular DI
- Bar Graph
- Line Graph
- Pie Chart

How To Interpret Data?

Data Interpretation requires students to make sense of numerical data that has been collected, analyzed, and presented. Given below are some of the steps that you can follow to interpret the data:

- Read the instruction carefully and all the variables given in the question. Understand the relation between the different variables first.
- Read the question carefully.
- Now, give the answer based on what you infer from the data.
- It is important that you know the basics of arithmetic because questions are based on that.
- Profit loss, Simple interest & Compound Interest are some of the examiner favourite topics for Data Interpretation.

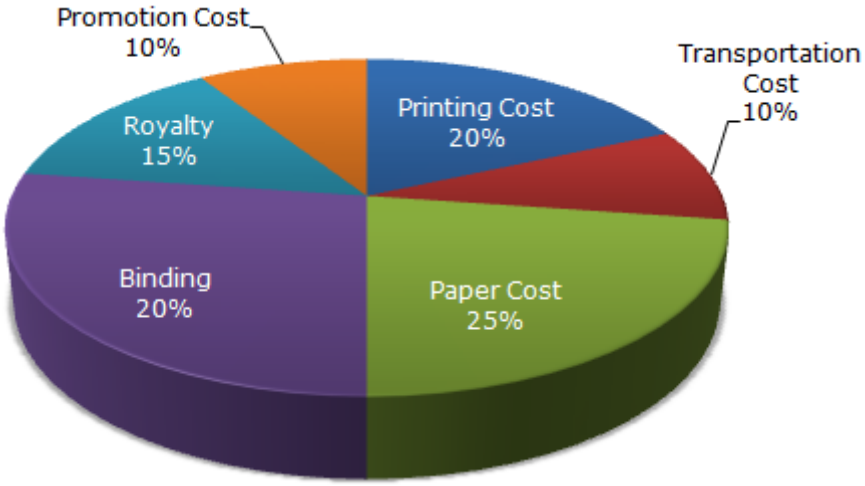
Why Data Interpretation Is Important

Data Interpretation is important because of the reasons given below:

- It helps in comparing two different variables.
- It is useful in establishing relations between two different entities.
- It enables you to merge two concepts into one.
- It helps the practical application in the workplace.
- It helps improve processes and identify problems.

Example: The following pie-chart shows the percentage distribution of the expenditure incurred in publishing a book. Study the pie-chart and the answer the questions based on it.

Various Expenditures (in percentage) Incurred in Publishing a Book



1. If for a certain quantity of books, the publisher has to pay Rs. 30,600 as printing cost, then what will be amount of royalty to be paid for these books?

A. Rs. 19,450

B. Rs. 21,200

C. Rs. 22,950

D. Rs. 26,150

2. What is the central angle of the sector corresponding to the expenditure incurred on Royalty?

A. 15°

B. 24°

C. 54°

D. 48°

3. The price of the book is marked 20% above the C.P. If the marked price of the book is Rs. 180, then what is the cost of the paper used in a single copy of the book?

A. Rs. 36

B. Rs. 37.50

C. Rs. 42

D. Rs. 44.25

4. If 5500 copies are published and the transportation cost on them amounts to Rs. 82500, then what should be the selling price of the book so that the publisher can earn a profit of 25%?

A. Rs. 187.50

B. Rs. 191.50

C. Rs. 175

D. Rs. 180

5. Royalty on the book is less than the printing cost by:

A. 5%

B. $33\frac{1}{5}\%$

C. 20%

D. 25%

1. **Answer:** Option **C**

Let the amount of Royalty to be paid for these books be Rs. *r*.

Then, 20 : 15 = 30600 : *r* $\Rightarrow$ *r* = Rs. $\left(\frac{30600 \times 15}{20}\right)$ = Rs. 22,950.

2. **Answer:** Option **C**

Central angle corresponding to Royalty = (15% of 360)°

$= \left(\frac{15}{100} \times 360\right)^{\circ}$

= 54°.

3. **Answer:** Option **B**

Clearly, marked price of the book = 120% of C.P.

Also, cost of paper = 25% of C.P

Let the cost of paper for a single book be Rs. *n*.

Then, 120 : 25 = 180 : *n* $\Rightarrow$ $n = \text{Rs.} \left(\frac{25 \times 180}{120}\right) = \text{Rs. } 37.50$.

4. **Answer:** Option **A**

For the publisher to earn a profit of 25%, S.P. = 125% of C.P.

Also Transportation Cost = 10% of C.P.

Let the S.P. of 5500 books be Rs. *x*.

Then, 10 : 125 = 82500 : *x* $\Rightarrow$ $x = \text{Rs.} \left(\frac{125 \times 82500}{10}\right) = \text{Rs. } 1031250.$

$\therefore$ S.P. of one book = Rs. $\left(\frac{1031250}{5500}\right) = \text{Rs. } 187.50$.

5.Answer: Option **D**

Printing Cost of book = 20% of C.P.

Royalty on book = 15% of C.P.

Difference = (20% of C.P.) - (15% of C.P) = 5% of C.P.

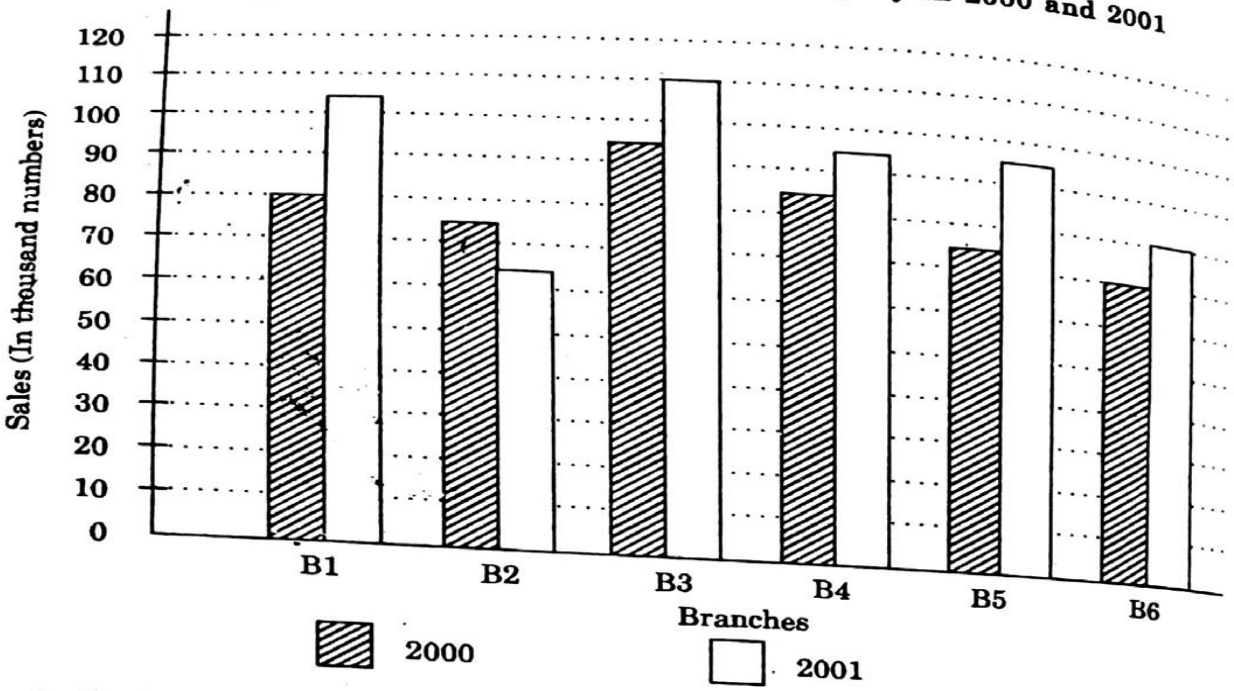
$\therefore$ Percentage difference $= \left(\frac{\text{Difference}}{\text{Printing Cost}} \times 100\right)\%$

$$= \left(\frac{5\% \text{ of C.P.}}{\text{Printing Cost}} \times 100\right)\% = 25\%.$$

Hence, required ratio = 5 : 6.

Ex. 2. *The bar-graph provided on next page gives the sales of books (in thousand numbers) from six branches of a publishing company during two consecutive years 2000 and 2001. Answer the questions based on this bar-graph.* **(Bank P.O. 2003)**

Sales of Books (in thousand numbers) from Six Branches — B1, B2, B3, B4, B5 and B6 of a Publishing Company in 2000 and 2001



1. Total sales of branches B1, B3 and B5 together for both the years (in thousand numbers) is :

(*a*) 250 (*b*) 310 (*c*) 435 (*d*) 560 (*e*) 585

2. Total sales of branch B6 for both the years is what percent of the total sales of branch B3 for both the years ?

(*a*) 68.54% (*b*) 71.11% (*c*) 73.17% (*d*) 75.55% (*e*) 77.26%

3. What is the average sale of all the branches (in thousand numbers) for the year 2000 ?

(*a*) 73 (*b*) 80 (*c*) 83 (*d*) 88 (*e*) 96

4. What is the ratio of the total sales of branch B2 for both years to the total sales of branch B4 for both years ?

(*a*) 2 : 3 (*b*) 3 : 5 (*c*) 4 : 5 (*d*) 5 : 7 (*e*) 7 : 9

5. What percent of the average sales of branches B1, B2 and B3 in 2001 is the average sales of branches B1, B3 and B6 in 2000 ?

(*a*) 75% (*b*) 77.5% (*c*) 82.5% (*d*) 85% (*e*) 87.5%

Sol. 1. (*d*) : Total sales of branches B1, B3 and B5 for both the years (in thousand numbers) = (80 + 105) + (95 + 110) + (75 + 95) = 560.

2. (*c*) : Required Percentage $= \left[\frac{(70+80)}{(95+110)} \times 100\right]\% = \left(\frac{150}{205} \times 100\right)\% = 73.17\%$.

3. (*b*) : Average sales of all the six branches (in thousand numbers) for the year 2000 $= \frac{1}{6} \times [80 + 75 + 95 + 85 + 75 + 70] = 80$.

4. (*e*) : Required ratio $= \frac{(75+65)}{(85+95)} = \frac{140}{180} = \frac{7}{9}$.

5. (e) : Average sales (in thousand numbers) of branches B1 B3 and B6 in 2000

$$= \frac{1}{3} \times (80 + 95 + 70) = \left(\frac{245}{3}\right).$$

Average sales (in thousand numbers) of branches B1, B2 and B3 in 2001

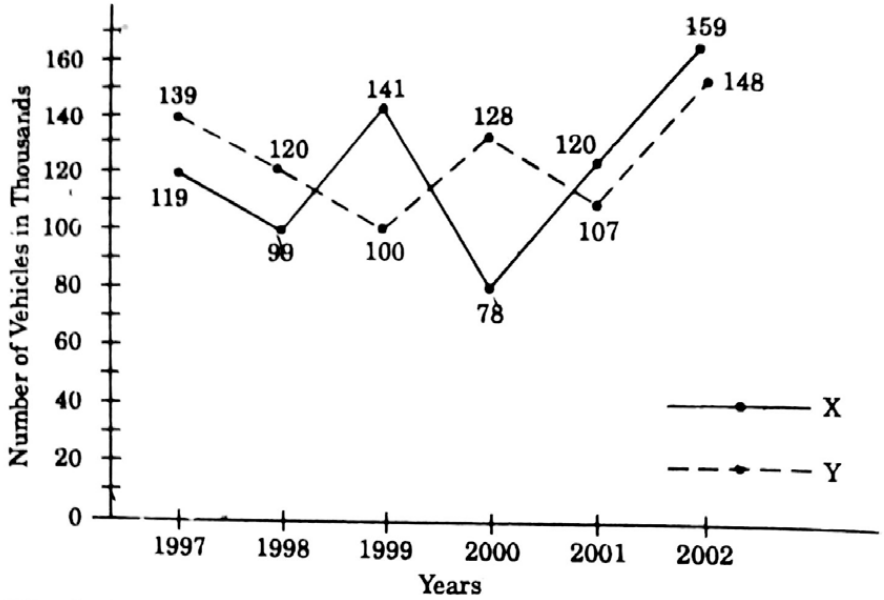
$$= \frac{1}{3} \times (105 + 65 + 110) = \left(\frac{280}{3}\right).$$

$$\therefore \quad \text{Required Percentage} = \left[\frac{\left(\frac{245}{3}\right)}{\left(\frac{280}{3}\right)} \times 100\right]\% = \left(\frac{245}{280} \times 100\right)\% = 87.5\%.$$

Ex. 3. *Study the following line-graph and answer the questions based on it.*

(R.B.I. 2003)

Number of Vehicles Manufactured by Two Companies over the Years
(Number in Thousands)



1. What is the difference between the total productions of the two Companies in the given years ?

   (*a*) 19000 (*b*) 22000 (*c*) 26000 (*d*) 28000 (*e*) 29000

2. What is the difference between the numbers of vehicles manufactured by Company Y in 2000 and 2001 ?

   (*a*) 50000 (*b*) 42000 (*c*) 33000 (*d*) 21000 (*e*) 13000

3. What is the average number of vehicles manufactured by Company X over the given period ? (rounded off to the nearest integer)

   (*a*) 119333 (*b*) 113666 (*c*) 112778 (*d*) 111223 (*e*) None of these

4. In which of the following years, the difference between the productions of Companies X and Y was the maximum among the given years ?

   (*a*) 1997 (*b*) 1998 (*c*) 1999 (*d*) 2000 (*e*) 2001

5. The production of Company Y in 2000 was approximately what percent of the production of Company X in the same year ?

   (*a*) 173 (*b*) 164 (*c*) 132 (*d*) 97 (*e*) 61

Sol. From the line-graph it is clear that the

(a) 175 (b) 164 (c) 132 (d) 97 (e) 61

Sol. From the line-graph it is clear that the productions of Company X in the years 1997, 1998, 1999, 2000, 2001 and 2002 are 119000, 99000, 141000, 78000, 120000 and 159000 respectively and those of Company Y are 139000, 120000, 100000, 128000, 107000 and 148000 respectively.

1. (*c*) : Total production of Company X from 1997 to 2002

$= 119000 + 99000 + 141000 + 78000 + 120000 + 159000 = 716000.$

and total production of Company Y from 1997 to 2002

$= 139000 + 120000 + 100000 + 128000 + 107000 + 148000 = 742000.$

Difference $= 742000 - 716000 = 26000.$

2. (*d*) : Required difference $= 128000 - 107000 = 21000.$

3. (*a*) : Average number of vehicles manufactured by Company X

$$= \frac{1}{6} \times (119000 + 99000 + 141000 + 78000 + 120000 + 159000) = 119333.$$

4. (*d*) : The difference between the productions of Companies X and Y in various years are :

For 1997 = (139000 − 119000) = 20000;

For 1998 = (120000 − 99000) = 21000;

For 1999 = (141000 − 100000) = 41000;

For 2000 = (128000 − 78000) = 50000;

For 2001 = (120000 − 107000) = 13000;

For 2002 = (159000 − 148000) = 11000.

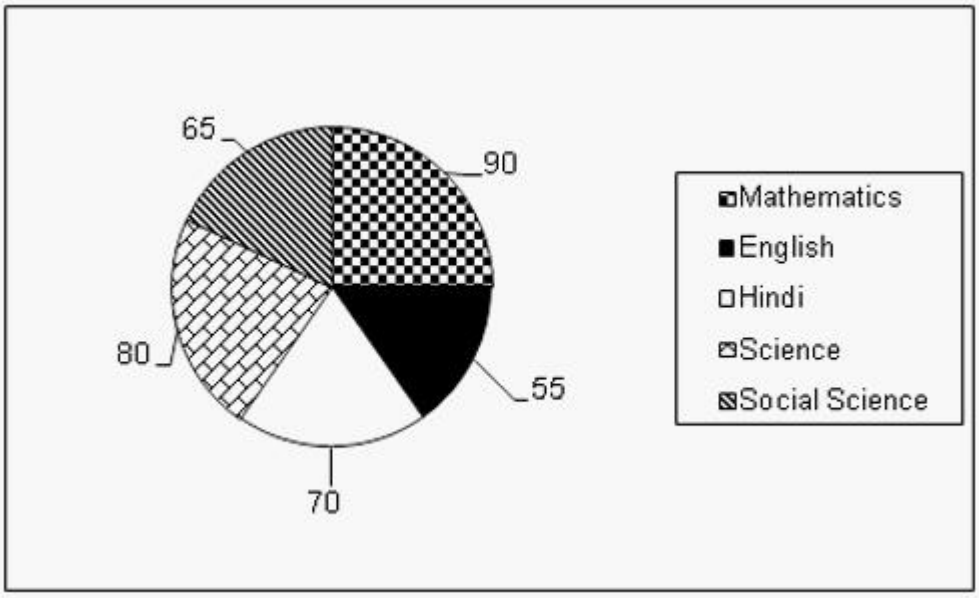
Clearly, maximum difference was in 2000

5. (*b*) : Required percentage $= \left(\frac{128000}{78000} \times 100\right)\% \approx 164\%.$

DIRECTIONS ***for questions 1-5:*** Refer to the pie chart given below and answer the questions that follow.

The given pie chart shows the marks scored by a student in different subjects- English, Hindi, Mathematics, Science and Social Science in an examination. The values given are in degrees.

Assumption: Total marks obtained in the examination are 900.



1. If the total marks were 3000, then marks in Mathematics would be

1. 800 2. 750 3. 850 4. 900

2. The Marks scored in English and Mathematics is less than the marks scored in Science and Hindi by

1. 5% 2. 4.33% 3. 3.33% 4. 6%

3. If the marks scored by the student are 137.5, then the subject is

1. English 2. Hindi 3. Mathematics 4. Science

4. Total marks scored in Social Science and English is

1. 400 2. 350 3. 500 4. 300

5. The difference of marks scored in Social Science and Science is

1. 37.5 2. 40 3. 20 4. 15

1. Marks obtained in Mathematics would be = 90 / 360 x 3000 = 750
2. **Answer Option: 3**
   Marks scored in English and Mathematics = (55 + 90) / 360 x 900 = 362.5 Marks scored in Hindi and Science = (70 + 80) / 360 x 900 = 375 Percent decrease = 12.5 / 375 x 100 = 3.33
3. **Answer Option: 1**
   Going by the options, marks scored in English = 55 /360 x 900 = 137.5
4. **Answer Option: 4**
   Angle of English sector is 55^0 and angle of Social Science sector is 65^0.So sum of angles is 120^0.Sum of marks =(120^0 x 900) / 360 = 300
5. **Answer Option: 1**
   Angle of Science sector is $80^0$and angle of Social Science sector is 65^0. So difference of angle is 15^0 .Difference of marks = (15^0 x 900) / 360^0 = 37.5